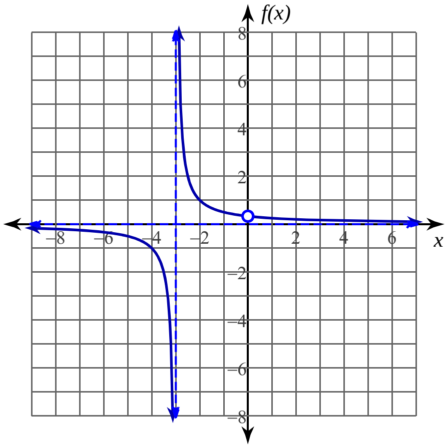
Name________________________________

Continuity

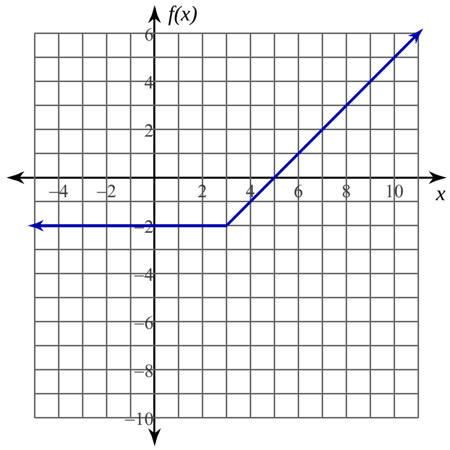
Date________________ Period____

Determine if each function is continuous at the given *x*-values. If not continuous, classify each discontinuity.

1) $f(x) = \dfrac{x}{x^2 + 3x}$; at $x = -3$ and $x = 0$



2) $f(x) = \begin{cases} -2, & x \le 3 \\ x - 5, & x > 3 \end{cases}$; at $x = 3$



3) $f(x) = \dfrac{x + 1}{x^2 + 2x + 2}$; at $x = -3$

4) $f(x) = \dfrac{x + 2}{x^2 - 4}$; at $x = -2$ and $x = 2$

5) $f(x) = \dfrac{x^2}{x + 1}$; at $x = -1$

6) $f(x) = \begin{cases} -2x, & x < 3 \\ -x^2 + 8x - 16, & x \ge 3 \end{cases}$; at $x = 3$

Determine if each function is continuous. If the function is not continuous, find the *x*-axis location of and classify each discontinuity.

7) $f(x) = -\dfrac{x}{2x^2 + 2x + 1}$

8) $f(x) = \dfrac{x}{x^2 + 6x + 9}$